# A Relation-Theoretic Result with Applications in Topological Algebra*

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**§0.** In this note we focus our attention on a result, which we call the Induced Function Theorem, intimately related to some work of M. L. Dubreil-Jacotin [5] and, to a lesser degree, of W. Simonsen [8]. Our statement and proof of the proposition are intended to reflect its relation-theoretic character, and most of the applications follow from a particularization of the relations involved in its statement. In general we ignore the application of the above-mentioned proposition to the derivation of numerous "isomorphism" theorems in algebra, particularly in group theory and semigroups, concentrating instead on its potential applications to the theory of machines, topologized as well as discrete. In particular, a recently proved result of Aczél and Wallace [1] is derived in a manner that we hope reflects the proposition's utility in this area and suggests further applications.

**§1.** A subset $R$ of $X \times Y$ will be called *a relation from $X$ into $Y$* if $X = Y$; this will be foreshortened to *a relation on $X$*. The inverse of the relation $R \subset X \times Y$ is the relation from $Y$ into $X$ denoted by $R^{-1}$ and defined by $R^{-1} = \{(y, x) | (x, y) \in R\}$. The relation on $X$ defined by $\{(x, x) | x \in X\}$ is called the *diagonal* of $X \times X$ and is denoted herein by diag $(X \times X)$.

If $R \subset X \times Y$ and if $A$ and $B$ are subsets of $X$ and $Y$ respectively, we denote by $AR$ the set $\{y | (a, y) \in R$ for some $a \in A\}$, while $RB$ is defined

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dually. We shall not distinguish between $xR$ and $\{x\}R$, preferring the former and depending upon the context to avoid ambiguity.

If $R \subset X \times Y$ and $S \subset Y \times Z$, the *composition* of $R$ and $S$, denoted in the sequel by $R \circ S$, is the relation from $X$ into $Z$ given by $R \circ S = \{(x, z) | (x, y) \in R$ and $(y, z) \in S$ for some $y \in Y\}$. In view of the preceding, the composition of $R$ and $S$ may be expressed by means of the set-theoretic statement $R \circ S = \bigcup\{Ry \times yS | y \in Y\}$.

Now if $R$ is a relation on $X$, the *transitive closure* of $R$, denoted by $\tau(R)$, is the relation on $X$ defined by

$$\tau(R) = R^{[1]} \cup R^{[2]} \cup R^{[3]} \cup \cdots \cup R^{[n]} \cup \cdots,$$

where $R = R^{[1]}$ and $R^{[n]} = R \circ R^{[n-1]}$ for $n \geq 2$. The relation $R$ on $X$ is *transitive* if $\tau(R) \subset R$. It should be observed that if $R$ is a reflexive (diag $(X \times X) \subset R$) and *symmetric* ($R = R^{-1}$) relation on $X$, then the transitive closure of $R$ is the minimal equivalence relation (reflexive, symmetric and transitive) containing $R$.

All of the necessary background material in topology may be found in Kelley [6] and reference to discrete semigroups is made *via* Clifford and Preston [4]. The expository dissertation of Paalman-de Miranda [7] is used as a general reference to the literature of topological semigroups.

**§2.** Basic to what follows is a lemma, the straightforward proof of which we omit.

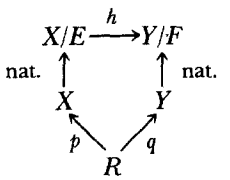
**LEMMA 1.** *Let $X$, $Y$ and $Z$ be all compact Hausdorff or all discrete, let $f: X \to Y$ and $g: X \to Z$ be continuous with $f$ onto, and let*

$$f(x) = f(x') \text{ imply } g(x) = g(x') \text{ for all } x, x' \in X,$$

*then there is a unique function $h: Y \to Z$, which is continuous, satisfying $g = hf$.*

If $R$ is a relation from $X$ into $Y$ we shall denote by $p$ and $q$ the first and second projections of $R$ into $X$ and $Y$ respectively.

**THEOREM 1.** (Induced Function Theorem). *Let $X$ and $Y$ be both compact Hausdorff or both discrete, let $R$ be a closed relation from $X$ into $Y$ and let $E$ and $F$ be closed equivalences on $X$ and $Y$ respectively. If $X = RY$ and if $R^{-1} \circ E \circ R \subset F$ then there exists a unique function $h: X/E \to Y/F$, which is continuous, for which the following diagram is analytic.*




*Moreover, if, in addition to the previous hypotheses, $Y = XR$ and $R \circ F \circ R^{-1} \subset E$, then $h$ is a homeomorphism.*

*Proof.* Denoting the natural maps by $\phi$ and $\psi$, that is, $\phi: X \to X/E$ and

$\psi$: $Y \to Y/F$, we note that in view of the assumption that $X = RY$ the composite map $\phi p$ is onto.

Now if $\phi p(x, y) = \phi p(x', y')$ for $(x, y)$ and $(x', y')$ in $R$, then $(x, x') \in E$ and hence

$$(y, y') = (y, x) \circ (x, x') \circ (x', y') \in R^{-1} \circ E \circ R \subset F.$$

This implies that $\psi(y) = \psi(y')$ or, equivalently,

$$\psi q(x, y) = \psi q(x', y').$$

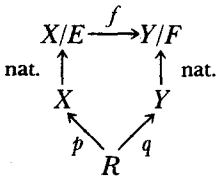
The existence of the map $h$ follows directly from the preceding lemma.

In a like manner, if in addition $Y = XR$ and $R \circ F \circ R^{-1} \subset E$, then there exists a unique map $g$: $Y/F \to X/E$ such that $h$ and $g$ are mutually inverse, therefore $h$ is a homeomorphism.

*Remark.* Actually it would have been sufficient to assume in the preceding that $Ex \cap RY \neq \square$ for each $x \in X$, a condition which is satisfied if $X = RY$.

We now turn our attention to a number of corollaries to the Induced Function Theorem (IFT), the first of these being a result contained implicitly in the paper of Simonsen [8].

**COROLLARY 1.** *Let $R$ be a relation from $X$ into $Y$ satisfying $X = RY$ and $Y = XR$. If $E$ and $F$ are the minimal equivalences containing $R \circ R^{-1}$ and $R^{-1} \circ R$ respectively, then there exists a unique one-to-one and onto map $f$: $X/E \to Y/F$ such that the diagram below is analytic.*




*Proof.* This follows directly from the induced function theorem after observing that if $X = RY$, the relation $R \circ R^{-1}$ is reflexive as well as symmetric, so that $E$ is precisely the transitive closure of $R \circ R^{-1}$. Similarly, $F$ is the transitive closure of $R^{-1} \circ R$ so that

$$R^{-1} \circ E \circ R = R^{-1} \circ \tau(R \circ R^{-1}) \circ R \subset \tau(R^{-1} \circ R) = F$$

and

$$R \circ F \circ R^{-1} = R \circ \tau(R^{-1} \circ R) \circ R^{-1} \subset \tau(R \circ R^{-1}) = E.$$

The following corollary is equivalent to the preceding, as can be seen from the way in which it is derived.

**COROLLARY 2.** *Let $R \subset X \times Y$ and $S \subset Y \times X$ be such that $X = RY$ and $Y = XS$. If $E$ and $F$ are the minimal equivalences containing $R \circ S$ and $S \circ R$ respectively, then there exists a one-to-one map $f$ of $X/E$ onto $Y/F$.*

*Proof.* Define the relation $T \subset X \times Y$ by

$$T = R \cup S^{-1}.$$

Note that $X = TY$ and $Y = XT$. The conclusion follows from Corollary 1 by observing that the minimal equivalence containing $T \circ T^{-1}$ is exactly the minimal equivalence containing $R \circ S$, and similarly for $T^{-1} \circ T$ and $S \circ R$.

Now

$$T \circ T^{-1} = (R \cup S^{-1}) \circ (R \cup S^{-1})^{-1} = (R \cup S^{-1}) \circ (R^{-1} \cup S)$$

$$= (R \circ R^{-1}) \cup (S^{-1} \circ R^{-1}) \cup (S^{-1} \circ S) \cup (R \circ S)$$

so that it is clear that the minimal equivalence containing $T \circ T^{-1}$, call it $E'$, contains $E$. To show the opposite inclusion, note first that $(R \circ S)$ and $(S^{-1} \circ R^{-1}) = (R \circ S)^{-1}$ are contained in $E$. Suppose $(x, a) \in (R \circ R^{-1})$, then for some $y \in Y$ we have $(x, y)$ and $(z, y)$ contained in $R$. Since $Y = SX$ there is a $w \in X$ such that $(y, w) \in S$, and thus $(x, w)$ and $(z, w)$ are elements of $R \circ S$. However this implies that $(x, w)$, $(z, w) \in E$ so that $(x, z) \in E$. The inclusion $S^{-1} \circ S \subset E$ is obtained in a similar way, completing the argument that $E = E'$.

A completely dual argument for $F$ establishes the desired conclusion.

Of course, Corollary 1 follows from Corollary 2 by taking $S = R^{-1}$.

We revert to topological assumptions to obtain the following.

**COROLLARY 3.** *Let $X$, $Y_1$ and $Y_2$ be all compact Hausdorff or all discrete. For $i = 1, 2$, let $R_i$ be a closed relation from $X$ into $Y_i$, and let $E_i$ and $F_i$ be closed equivalences on $X$ and $Y_i$, respectively satisfying*

$$X = R_i Y_i,\ Y_i = X R_i,\ R_i^{-1} \circ E_i \circ R_i \subset F_i$$

*and $R_i \circ F_i \circ R_i^{-1} \subset E_i$. Suppose furthermore that $R_1^{-1} \circ E_2 \circ R_1 \subset F_1$ and $R_2^{-1} \circ E_1 \circ R_2 \subset F_2$, then $Y_1/F_1$ and $Y_2/F_2$ are topologically equivalent.*

*Proof.* Define the relation $P$ from $Y_1$ into $Y_2$ by $P = R_1^{-1} \circ R_2$. If $y \in Y_1$, then there is an $x_0 \in X$ such that $(x_0, y) \in R_1$. Now $X = R_2 Y_2$ implies there exists a $y' \in Y_2$ such that $(x_0, y') \in R_2$. Thus

$$(y, y') \in R_1^{-1} \circ R_2 = P,$$

so that $y \in PY_2$. In a similar way we may show that $Y_2 = Y_1 P$.

Considering $P^{-1} \circ F_1 \circ P$, we see that

$$P^{-1} \circ F_1 \circ P = (R_1^{-1} \circ R_2)^{-1} \circ F_1 \circ (R_1^{-1} \circ R_2)$$

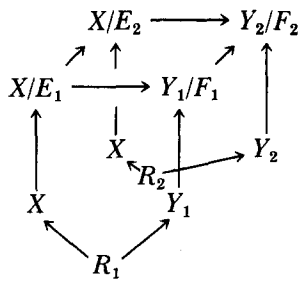
$$= R_2^{-1} \circ (R_1 \circ F_1 \circ R_1^{-1}) \circ R_2$$

$$\subset R_2^{-1} \circ E_1 \circ R_2 \subset F_2$$

and, similarly, $P \circ F_2 \circ P^{-1} \subset F_1$. Thus all of the hypotheses of (IFT) are satisfied so that there is a homeomorphism from $Y_1/F_1$ onto $Y_2/F_2$.

*Remark.* It may be of some interest to note the diagram corresponding to Corollary 3. With $\overset{t}{=}$ denoting topological equivalence, the hypotheses of the corollary guarantee that $X/E_1 \overset{t}{=} Y/F_1$, $X/E_2 \overset{t}{=} Y/F_2$ and, as we have

seen, $Y/F_1 \overset{t}{=} Y/F_2$. Thus all of the nodes on the upper "face" of the diagram are topologically equivalent.




**§3.** We focus our attention on applications of the preceding results. As mentioned earlier our principal concern here is with applications to topological algebra, more specifically to topologized "machines".

A *recursion* is a continuous function

$$T \times X \to X$$

such that $T$ and $X$ are nonvoid Hausdorff spaces. The value of the function at $(t, x)$ is denoted by $tx$. If $T$ is a groupoid, its multiplication will also be denoted by juxtaposition.

A recursion is called discrete [compact] if both $T$ and $X$ are discrete [compact].

If, in the recursion $T \times X \to X$, $T' \subset T$ and $X' \subset X$, then

$$T'X' = \{tx |\ t \in T' \text{ and } x \in X'\}$$

and $T'x$ is used in place of $T'\{x\}$.

**THEOREM 2.** *Let $T \times X \to X$ and $T \times Y \to Y$ be both compact or both discrete recursions. Suppose that $Tx_0 = X$ and $Ty_0 = Y$ for some $x_0 \in X$ and $y_0 \in Y$. If, for all $t, t' \in T$,*

$$(*) \qquad tx_0 = t'x_0 \text{ if and only if } ty_0 = t'y_0,$$

*then $X \overset{t}{=} Y$.*

*Proof.* Define $R_1 \subset T \times X$ and $R_2 \subset T \times Y$ by $R_1 = \{(t, x) |\ tx_0 = x\}$ and $R_2 = \{(t, y) |\ ty_0 = y\}$. Let $E_1 = R_1 \circ R_1^{-1}$, $F_1 = R_1^{-1} \circ R_1$, $E_2 = R_2 \circ R_2^{-1}$ and $F_2 = R_2^{-1} \circ R_2$. If we observe that diag $(X \times X) = F_1$, diag $(Y \times Y) = F_2$ and that (*) implies $E_1 = E_2$, it is immediate that all of the hypotheses of Corollary 3 are satisfied so that

$$X \overset{t}{=} X/\text{diag } (X \times X) \overset{t}{=} Y/\text{diag } (Y \times Y) \overset{t}{=} Y.$$

As a second application of (IFT) we consider the case in which, rather than having a recursion, we have a semigroup $T$, a space $X$ and a *relation* $R$ from $T \times X$ into $X$. Given a congruence $E$ on $T$ and an equivalence $F$ on $X$ we are interested in relation-theoretic conditions under which we may generate a *topological machine*, $T/E \times X/F \to X/F$.

Recall that a *semigroup* (topological) is a nonvoid Hausdorff space with a continuous associative multiplication. If $T$ is a semigroup and if $E$ is an equivalence on $T$, then $E$ is called a congruence if $(t_1, t_2) \in E$ implies that $(tt_1, tt_2)$ and $(t_1t, t_2t)$ are contained in $E$ for each $t \in T$.

By *topological machine* we mean a continuous function

$$T \times X \to X$$

such that $T$ is a semigroup, $X$ is a nonvoid Hausdorff space and

$$t_1(t_2x) = (t_1t_2)x \text{ for all } t_1, t_2 \in T \text{ and all } x \in X.$$

If $E$ is an equivalence on $T$ and $F$ is an equivalence on $X$, then we define the product of $E$ and $F$, denoted by $E \times F$, to be the relation on $T \times X$ defined by $((t, x), (t', x')) \in E \times F$ if and only if $(t, t') \in E$ and $(x, x') \in F$. As noted in Bourbaki [3, p. 122], $E \times F$ is an equivalence relation on $T \times X$.

*Throughout the remainder of this section we use $\phi$, $\psi$ and $\eta$ to denote the natural maps $\phi$: $T \to T/E$, $\psi$: $X \to X/F$ and $\eta$: $T \times X \to T \times X/E \times F$.* As a direct consequent of Lemma 1, we have:

**LEMMA 2.** *If $T$ and $X$ are both compact Hausdorff or both discrete and if $E$ and $F$ are closed equivalences on $T$ and $X$ respectively, then there exists a unique homeomorphism $h$: $T/E \times X/F \to T \times X/E \times F$ for which $h(\phi \times \psi) = \eta$.*

**THEOREM 3.** *Let $T$ and $X$ be both compact Hausdorff or both discrete and let $T$ be, in addition, a semigroup. If $R$ is a closed relation from $T \times X$ into $X$, $E$ is a closed congruence on $T$ and $F$ is a closed equivalence satisfying*

(i) $$T \times X = RX$$

(ii) $$R^{-1} \circ (E \times F) \circ R \subset F$$

(iii) $$(t_1t_2, x)R \times (t_1, z)R \subset F$$

*for all $t_1, t_2 \in T$ and all $z \in (t_2, x)R$, then there exists a unique continuous function $T/E \times X/F \to X/F$ for which $T/E \times X/F \to X/F$ is a topological machine and the diagram below is analytic.*

$$\begin{array}{ccc} T/E \times X/F & \longrightarrow & X/F \\ {\scriptstyle(\phi \times \psi)}\uparrow & & \uparrow{\scriptstyle\psi} \\ T \times X & & X \\ & {\scriptstyle p}\nwarrow \quad \nearrow{\scriptstyle q} & \\ & R & \end{array}$$

*Proof.* Using (i), (ii), and Theorem 1, we have that there exists a unique continuous function $h$: $T \times X/E \times F \to X/F$ for which the diagram

$$\begin{array}{ccc} T \times X/E \times F & \xrightarrow{h} & X/F \\ {\scriptstyle\eta}\uparrow & & \uparrow{\scriptstyle\psi} \\ T \times X & & X \\ & {\scriptstyle p}\nwarrow \quad \nearrow{\scriptstyle q} & \\ & R & \end{array}$$

is analytic. By Lemma 2, there is a unique homeomorphism $h'$: $T/E \times X/F \to T \times X/E \times F$ for which $h'(\phi \times \psi) = \eta$. Let $f$: $T/E \times X/F \to X/F$ be defined by $f = hh'$.

Since $T/E$ is a semigroup we need only show that

$$f(\phi(t_1)\phi(t_2), \psi(x)) = f(\phi(t_1), f(\phi(t_2), \psi(x)))$$

for all $t_1, t_2 \in T$ and for each $x \in X$.

Now $f(\phi(t_1)\phi(t_2), \psi(x)) = f(\phi(t_1t_2), \psi(x)) = f(\phi \times \psi)(t_1t_2, x)$. Since $T \times X = RX$, there is an $x_0 \in X$ such that $((t_1t_2, x), x_0) \in R$, so that

$$f(\phi(t_1)\phi(t_2), \psi(x)) = f(\phi \times \psi)p((t_1t_2, x), x_0) = \psi q((t_1t_2, x), x_0) = \psi(x_0).$$

But $f(\phi(t_1), f(\phi(t_2), \psi(x))) = f(\phi(t_1), f(\phi \times \psi)(t_2, x))$, and since $T \times X = RX$ we may find $x_1, x_2 \in X$ such that $((t_2, x), x_1) \in R$ and $((t_1, x_1), x_2) \in R$. Then

$$\begin{aligned} f(\phi(t_1), f(\phi(t_2), \psi(x))) &= f(\phi(t_1), f(\phi \times \psi)p((t_2, x), x_1)) \\ &= f(\phi \times \psi)p((t_1, x_1), x_2) = \psi(x_2). \end{aligned}$$

However by (iii), $(x_0, x_2) \in (t_1t_2, x)R \times (t_1, x_1)R$, where $x_1 \in (t_2, x)R$, and hence $\psi(x_0) = \psi(x_2)$, completing the proof.

A special case of the preceding was proved in [2] where one begins with a compact or discrete topological machine $T \times X \to X$, a closed congruence $E$ on $T$ and a closed equivalence $F$ on $X$ satisfying $EF \subset F$ (the multiplication being coordinate-wise). There, of course, the relation $R$ is taken to be the "graph" of the function $T \times X \to X$, that is,

$$R = \{((t, x), y) \mid tx = y\}.$$

Our final formal application of (IFT) involves the derivation of a recently proved result of Aczél and Wallace [1] which concerns itself with conditions on a recursion which force the state space to admit a particular algebraic structure.

**THEOREM 4.** (Aczél-Wallace). *Let $T \times X \to X$ be a compact or discrete recursion satisfying* (i) *$Tz = X$ for some $z \in X$ and* (ii) *$t_1(t_2x) = t_2(t_1x)$ for all $t_1, t_2 \in T$ and for each $x \in X$. Then $X$ is a commutative semigroup under an operation $*$ with $z$ as unit such that $tx = (tz) * x$ for each $t \in T$ and each $x \in X$.*

*Proof.* Defining the relation $R$ from $T$ into $X$ by $(t, x) \in R$ if and only if $tz = x$, we see that $T = RX$ and $X = TR$. Now $E = R \circ R^{-1} = \{(t, t') \mid tz = t'z\}$ is an equivalence. Similarly, $F = R^{-1} \circ R = \text{diag}\,(X \times X)$, so that by Corollary 1 to Theorem 1 there exists a unique one-to-one and onto function $f$: $T/E \to X/\text{diag}\,(X \times X)$ for which $f\phi p = \psi q$.

On the topological side, the relation $R$ being closed, it follows that $E$ is also, so that $f$ is a homeomorphism. Denote the inverse of $f$ by $g$.

To complete the proof we show that $T/E$ admits a continuous, commutative, associative multiplication $\circ$ having $g\psi(z)$ as unit. Observing that $(t, t') \in E$ implies $tx = t'x$ for all $x \in X$, by Lemma 1 there is a unique continuous map $\circ$: $T/E \times T/E \to T/E$ such that the diagram below is analytic.

$$\begin{array}{ccc} T/E \times T/E & \xrightarrow{\circ} & T/E \\ {\scriptstyle(\phi \times g\psi)}\uparrow & & \uparrow{\scriptstyle g\psi} \\ T \times X & \longrightarrow & X \end{array}$$

Since $(t, tz) \in R$ for each $t \in T$, we have $g\psi(tz) = \phi(t)$. Using this we note that

$$\begin{aligned} \phi(t_1) \circ \phi(t_2) &= \phi(t_1) \circ g\psi(t_2 z) = g\psi(t_1(t_2 z)) \\ &= g\psi(t_2(t_1 z)) = \phi(t_2) \circ \phi(t_1), \end{aligned}$$

that is, the operation $\circ$ is commutative. Now

$$\begin{aligned} \phi(t_1) \circ (\phi(t_2) \circ \phi(t_3)) &= \phi(t_1) \circ g\psi(t_2(t_3 z)) \\ &= g(t_1(t_2(t_3 z))). \end{aligned}$$

Similarly, $(\phi(t_1) \circ \phi(t_2)) \circ \phi(t_3) = g(t_3(t_1(t_2 z)))$ from which it follows, since $t(t'x) = t'(tx)$ for all $t, t' \in T$ and all $x \in X$, that

$$\phi(t_1) \circ (\phi(t_2) \circ \phi(t_3)) = (\phi(t_1) \circ \phi(t_2)) \circ \phi(t_3).$$

Let $t_0 \in T$ be such that $t_0 z = z$; then for any $t \in T$, $\phi(t) \circ \phi(t_0) = g\psi(t(t_0 z)) = g\psi(tz) = \phi(t)$, so that $\phi(t_0) = g\psi(t_0 z) = g\psi(z)$ is a *unit* for $T/E$.

We have restricted ourselves in the preceding to applications concerning, primarily, abstract machines, topologized as well as discrete. It is clear that the propositions considered may be applied to generate many of the classical isomorphism theorems in algebra. In addition the topologized and relativized version [9] of the Schützenberger-Preston Theorem can be obtained directly from Corollary 3, but the introduction of the notation and concepts to verify this formally would have led us too far afield from the original intent of this note.

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